

ISLAND FALLS, SK, Canada

WMO: 713310

Lat: 55.530N

Lon: 102.350W

Elev: 976

StdP: 14.19

Time Zone: -6.00 (W06)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.6	-28.5	-41.3	0.5	-33.5	-35.7	0.7	-28.3	12.7	-0.1	11.6	-0.3	1.6	240	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	81.3	62.7	78.3	61.1	75.3	59.6	65.3	77.0	63.6	74.2	61.8	71.8	6.3	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.9	82.8	70.0	59.2	77.9	68.2	57.6	73.4	66.6	30.8	77.7	29.5	74.0	28.2	71.9	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.5	11.7	10.1	DB	-39.9	87.0	5.0	3.0	-43.5	89.2	-46.5	91.0	-49.3	92.7	-52.9	94.9	(4)
(5)			WB	-40.0	68.5	4.9	1.7	-43.4	69.8	-46.3	70.8	-49.0	71.7	-52.5	73.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	30.8	-2.6	-1.2	13.6	31.4	45.6	57.1	62.6	59.9	49.8	35.7	15.9	-0.1		
	DBStd	26.04	14.09	13.10	13.69	11.85	8.59	6.99	5.47	5.67	8.12	7.76	12.88	14.53		
	HDD50	8091	1630	1435	1127	562	189	20	0	3	101	449	1023	1552		
	HDD65	12544	2095	1855	1592	1009	601	247	112	176	458	909	1473	2017		
	CDD50	1089	0	0	0	3	53	233	391	309	96	4	0	0		
	CDD65	69	0	0	0	0	1	11	37	17	3	0	0	0		
	CDH74	903	0	0	0	0	46	176	429	201	51	0	0	0		
	CDH80	139	0	0	0	0	6	22	82	22	7	0	0	0		
	WSAvg	4.6	4.2	4.0	4.4	4.9	5.2	4.5	4.5	4.8	4.8	4.9	4.5	4.0		
	PrecAvg	20.4	0.9	0.6	0.7	0.8	1.8	2.9	3.4	2.9	2.8	1.5	1.1	1.1		
(15) Precipitation	PrecMax	26.3	1.5	1.5	1.7	1.7	5.2	8.0	7.4	5.4	7.6	2.6	2.8	2.1		
	PrecMin	13.4	0.2	0.0	0.1	0.0	0.4	0.8	1.1	0.7	0.4	0.3	0.3	0.3		
	PrecStd	2.8	0.4	0.4	0.4	0.4	1.0	1.5	1.4	1.3	1.7	0.7	0.6	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.0	35.7	51.1	67.3	79.3	82.8	86.3	82.8	79.4	63.7	44.3	32.3		
		MCWB	30.3	32.3	40.2	48.7	55.9	62.7	64.5	64.2	61.5	50.8	38.1	29.9		
	2%	DB	27.6	30.1	44.0	60.2	73.7	78.8	82.0	79.2	73.9	56.5	39.6	27.5		
		MCWB	25.8	27.2	35.2	45.5	53.9	59.6	63.9	63.1	59.3	47.4	35.4	25.8		
	5%	DB	22.4	24.6	38.5	54.4	69.4	75.4	78.8	75.9	68.8	51.6	35.6	23.6		
		MCWB	21.0	22.0	32.5	42.1	52.1	58.2	61.9	61.3	56.9	45.6	32.9	22.2		
	10%	DB	18.3	19.5	33.9	49.6	63.9	72.0	75.5	72.1	63.8	47.5	32.4	20.6		
		MCWB	17.1	17.7	29.8	39.1	49.2	56.8	60.8	59.6	54.7	42.7	30.2	19.6		
	0.4%	WB	30.4	32.5	40.2	50.0	58.3	65.8	67.7	67.2	63.9	52.9	39.1	30.4		
		MCDB	32.6	35.4	50.4	65.3	74.8	77.5	79.3	78.9	76.3	60.0	42.7	32.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	25.9	27.0	35.9	45.8	55.2	63.2	65.7	64.6	60.4	48.5	35.6	25.9		
		MCDB	27.5	29.7	42.8	59.4	70.2	73.9	77.9	75.3	71.5	55.0	38.7	27.3		
	5%	WB	21.3	22.2	33.0	42.7	52.7	60.9	64.1	62.7	58.1	45.9	32.9	22.3		
		MCDB	22.3	24.4	37.8	53.5	67.0	70.3	75.0	72.5	67.5	51.5	35.6	23.6		
	10%	WB	17.2	17.7	30.0	39.6	50.1	58.7	62.3	60.9	55.1	43.0	30.1	19.5		
		MCDB	18.3	19.6	33.9	48.4	63.5	68.3	72.7	69.8	63.0	47.3	32.3	20.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.4	21.6	24.0	23.1	23.6	21.6	20.6	18.7	18.1	12.3	12.4	14.4		
		MCDBR	17.9	22.8	25.1	29.6	34.6	28.2	26.7	26.1	26.1	20.6	14.9	14.6		
		MCWBR	16.9	20.1	17.9	17.1	17.3	12.7	12.2	12.6	13.7	13.1	11.4	13.7		
		MCDBR	17.8	22.2	22.3	28.1	30.2	21.9	22.7	21.9	23.8	18.6	13.9	15.0		
	5% WB	MCWBR	16.8	19.7	16.0	16.7	15.5	11.1	11.7	11.8	13.2	12.1	10.9	14.1		
(40) Clear-Sky Solar Irradiance	taub		0.232	0.249	0.268	0.319	0.330	0.367	0.393	0.359	0.319	0.287	0.242	0.216		
	taud		2.283	2.316	2.332	2.394	2.487	2.409	2.366	2.471	2.566	2.576	2.419	2.265		
	Ebn at Noon		231	270	290	288	290	279	268	272	271	256	229	210		
	Edh at Noon		16	23	29	32	31	35	36	30	24	18	14	13		
(44) All-Sky Solar Radiation	RadAvg		248	591	1093	1587	1756	1794	1735	1415	960	501	232	148		
	RadStd		29	46	84	108	175	151	111	117	111	72	28	21		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page